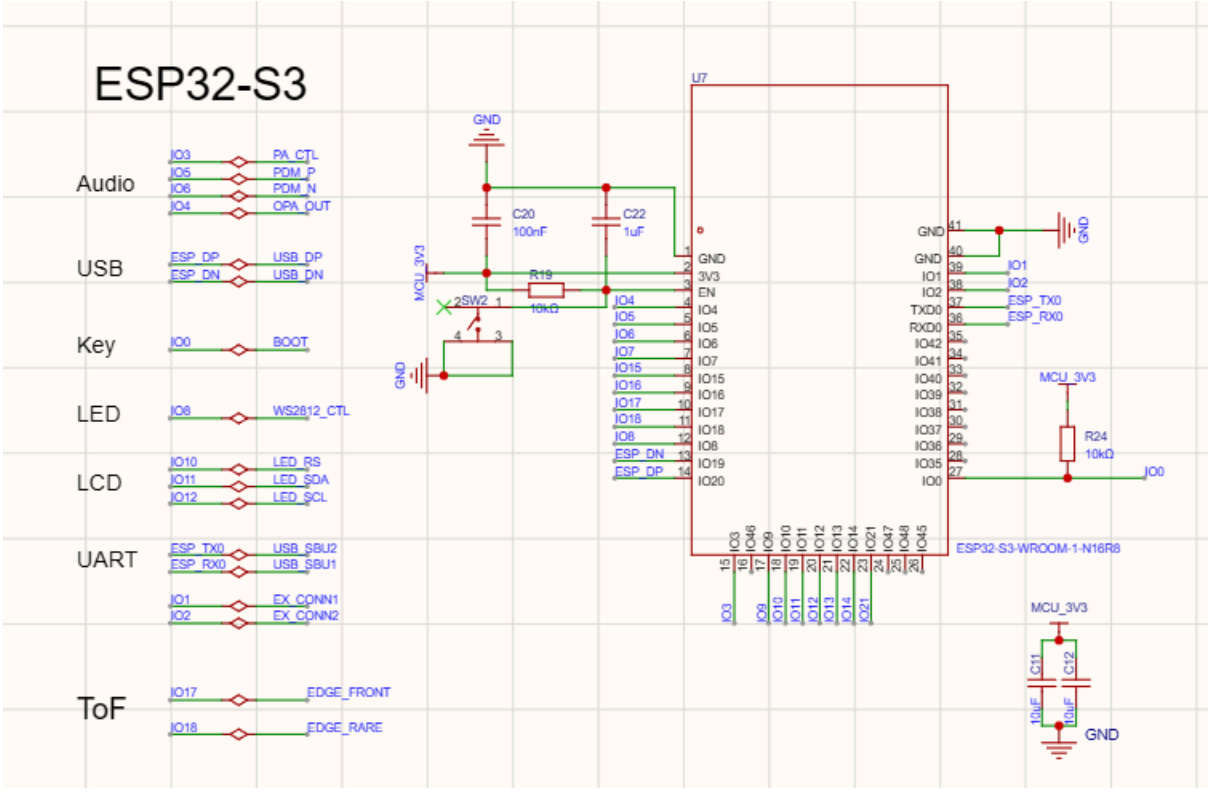
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Core Controller and System Interconnect



This schematic diagram illustrates the central processing hub of the robot, based on the **ESP32-S3-WROOM-1-N16R8** module. It serves as the "brain" of the system, managing all high-level tasks including dual-core processing, Wi-Fi/Bluetooth connectivity, and the orchestration of various subsystems such as Audio, USB communication, LCD interface, and sensor data acquisition.

2. Design Rationale

- **Module Selection**: The **ESP32-S3** variant was chosen specifically for its integrated **AI acceleration instructions** and its **N16R8 configuration** (16MB Flash / 8MB PSRAM). This memory overhead is critical for handling the image frame buffers required for the computer vision tasks and the complex data structures involved in the Cloud AI interaction.
- **Power Stability & Decoupling**: Multiple decoupling capacitors (**C20, C22, C11, C12**) are placed near the VCC pins. The combination of **100nF** (for high-frequency noise) and **1uF/10uF** (for bulk energy storage) ensures that the SoC remains stable during transient power spikes caused by Wi-Fi bursts or motor startups.
- **Bootstrap & Reset Logic**:
    - **BOOT (IO0)** and **EN (Reset)** circuits are designed with **10kΩ pull-up resistors (R24, R19)** to ensure the chip defaults to the correct execution mode.
    - The manual **BOOT button (SW2)** allows for easy entry into firmware download mode, facilitating rapid iterative development.

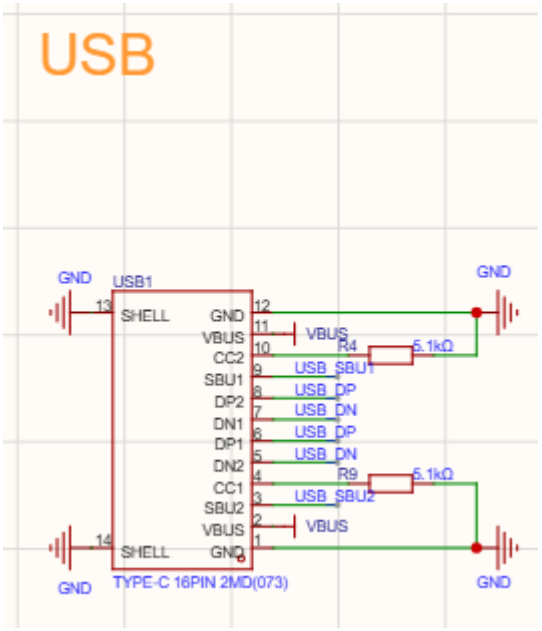
- **Peripheral Mapping**:
    - **USB Native Interface**: Utilizes **IO19/IO20** for direct USB communication, which simplifies debugging and eliminates the need for an external USB-to-UART chip.
    - **Audio PDM**: Dedicated pins (**IO5, IO6**) are assigned for differential PDM output, matching the requirements of the downstream power amplifier (PA) circuit.
    - **LCD & LED Control**: High-speed SPI pins and dedicated PWM-capable GPIOs (like **IO8** for WS2812B) are selected to ensure low-latency visual feedback.

3. Expected Outcome

This configuration provides a robust and scalable foundation for the robot's hardware. By utilizing the native USB and ample PSRAM, the design achieves high-speed data throughput for both programming and real-time sensor processing. The careful management of decoupling and strapping pins minimizes the risk of "brown-out" resets or boot failures, ensuring the system remains operational under the diverse electrical loads present in a mobile robotics platform.

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USB-C Communication and Power Input Interface



This schematic section details the **USB Type-C (16-pin)** connector interface. It serves as the primary gateway for **firmware deployment**, **serial debugging**, and **5V system power input**. By utilizing the native USB capabilities of the ESP32-S3, this interface allows the robot to be programmed and powered through a single, industry-standard cable.

2. Design Rationale

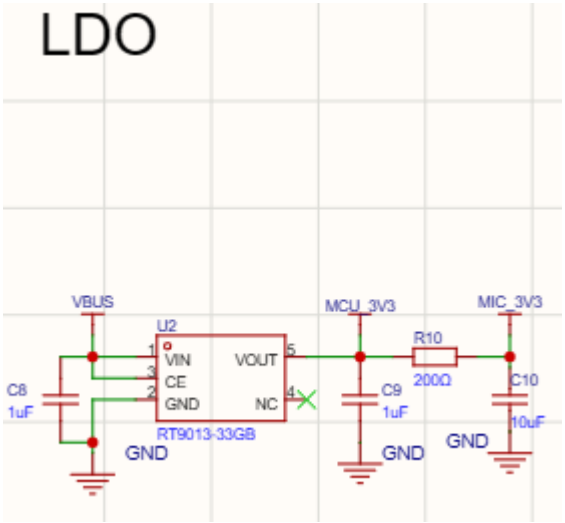
- **Universal Compatibility (CC Logic)**: The inclusion of two **5.1kΩ pull-down resistors (R4, R9)** on the **CC1** and **CC2** pins is a critical design choice. These resistors signal to a USB-C Power Delivery (PD) source or an E-marked cable that the robot is a "Sink" device (power consumer), ensuring that the upstream source provides the necessary 5V on the VBUS line.
- **Signal Integrity**: The differential data pairs (**USB_DP** and **USB_DN**) are mapped to pins 5 through 8. In the subsequent PCB layout phase, these will be routed as a **90Ω differential pair** to maintain signal impedance and prevent data corruption during high-speed transfers.
- **Reversible Connectivity**: The pinout configuration is symmetrical, supporting the reversible nature of the USB-C standard. This enhances user convenience and reduces mechanical wear on the port.
- **Mechanical and Electrical Grounding**: The **SHELL** pins (13, 14) are tied directly to **GND**. This provides structural stability to the connector and creates a low-impedance path for Electrostatic Discharge (ESD), protecting the sensitive internal SoC from potential high-voltage transients during plug-in events.

3. Expected Outcome

The implementation of this USB-C interface ensures seamless integration with modern development environments. It provides a reliable **12Mbps (Full Speed)** data link to the ESP32-S3's internal USB-Serial/JTAG controller, enabling real-time monitoring of the robot's "brain" without the need for external bridge chips. Furthermore, the standardized power

negotiation ensures the robot can be safely powered by a wide variety of USB chargers and laptop ports.

Power Regulation and Low-Noise Supply Network



1. Functional Overview

This schematic section highlights the **Low-Dropout (LDO) Regulator** circuit, which is responsible for stepping down the **5V VBUS** (from USB) to a stable **3.3V** required by the system's digital and analog components. It utilizes the **RT9013-33GB** to provide a clean, regulated power rail, with specific filtering tailored for noise-sensitive peripherals like the microphone.

2. Design Rationale

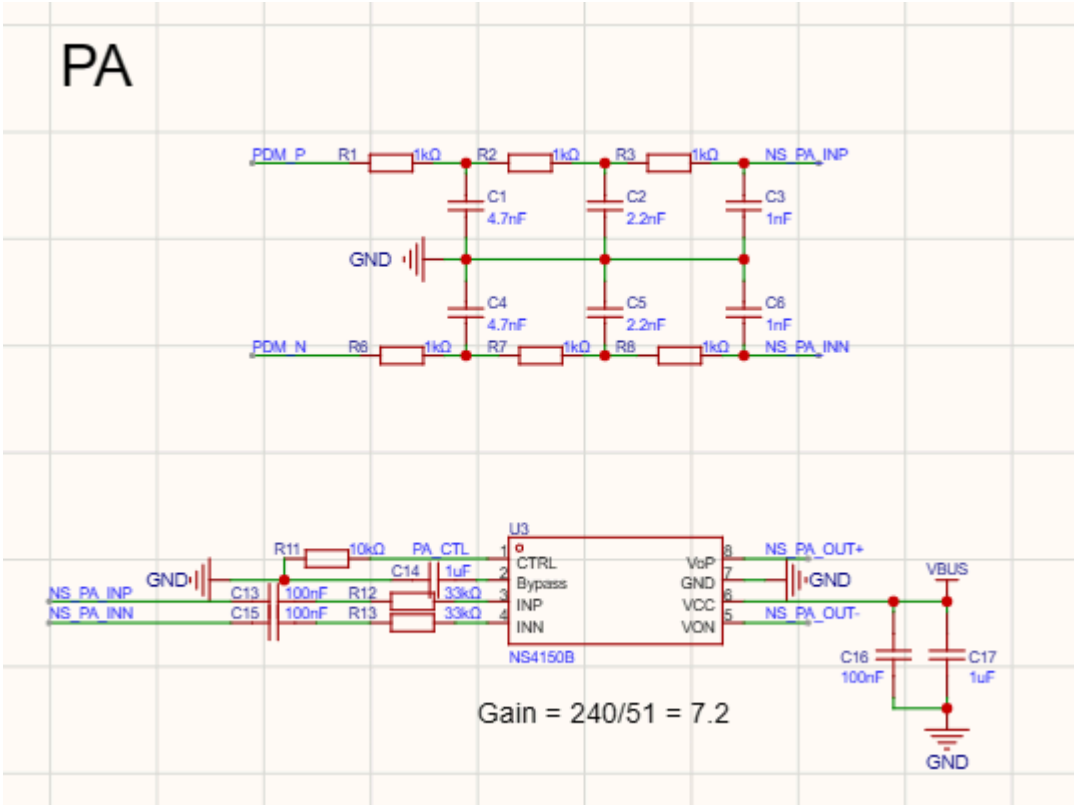
- **Regulator Selection**: The **RT9013** was selected for its high **Power Supply Rejection Ratio (PSRR)** and low quiescent current. This ensures that high-frequency noise from the USB power source is effectively filtered out before reaching the MCU.
- **Input/Output Filtering**:
    - **C8 (1uF)** acts as the input reservoir, stabilizing the 5V line and providing local energy storage for the regulator.
    - **C9 (1uF)** provides primary output stability for the **MCU_3V3** rail, ensuring the ESP32-S3 receives a steady voltage during varying logic states.
- **Analog Noise Isolation (RC Filter)**:
    - A critical design feature is the **RC Low-Pass Filter** composed of **R10 (200Ω)** and **C10 (10uF)**.
    - This secondary stage creates the **MIC_3V3** rail. By isolating the microphone's power supply from the main digital rail, we prevent high-frequency digital switching noise from the MCU from coupling into the sensitive audio pre-amplifier stage.

- **Enable Logic**: The **CE (Chip Enable)** pin is tied directly to **VIN**, ensuring the 3.3V rail is active as soon as USB power is applied, which is standard for this bus-powered application.

3. Expected Outcome

This power architecture provides a dual-benefit: a stable voltage source for the main controller and a highly filtered, "clean" rail for the audio sensors. By implementing the RC filter for the microphone supply, the design significantly reduces the noise floor of the recorded audio, which is essential for the reliability of the downstream **Cloud AI voice recognition** and **ASR** tasks. The overall thermal performance remains excellent due to the efficient selection of the LDO for the intended current draw.

Audio Power Amplifier and Differential Filtering



1. Functional Overview

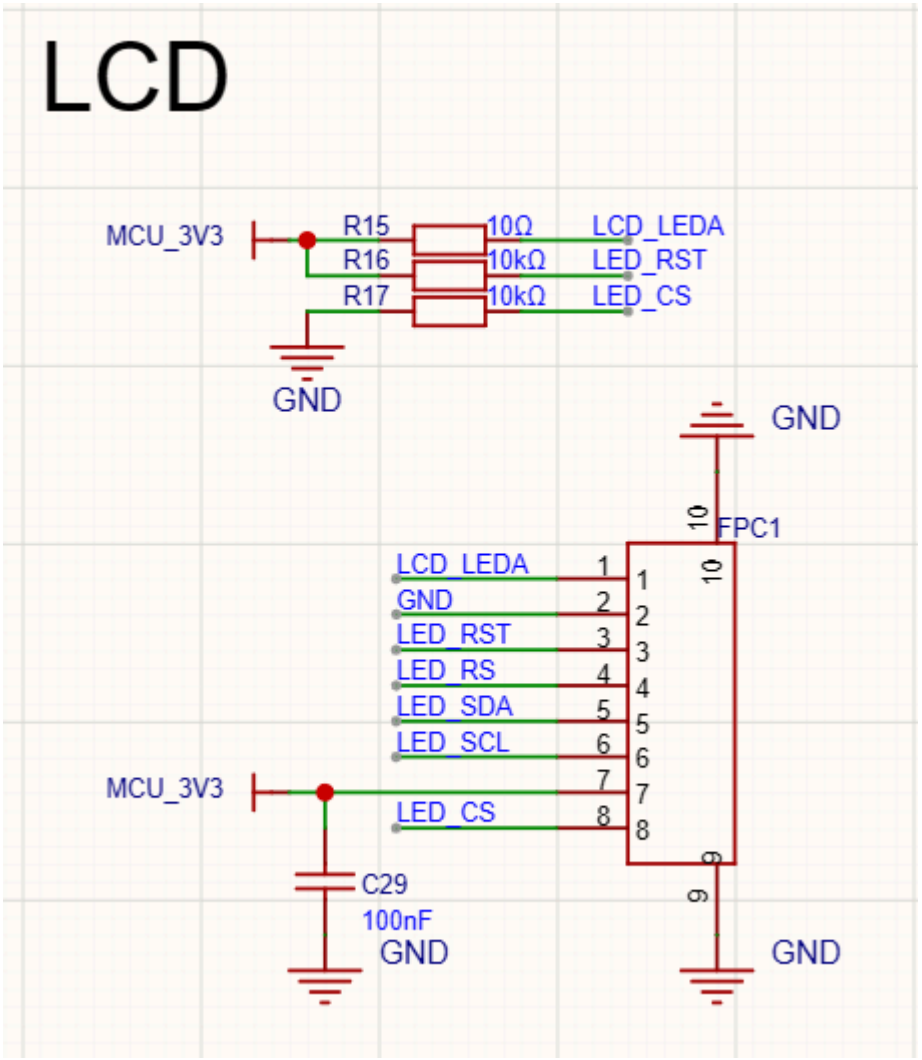
This schematic section represents the robot's **Audio Power Amplifier (PA)** subsystem. It is designed to convert high-frequency **Pulse Density Modulation (PDM)** digital signals from the ESP32-S3 into audible analog signals to drive the speaker. The circuit features a multi-stage passive filter followed by an **NS4150B** differential amplifier, ensuring high-fidelity audio playback for the robot's voice and interaction sounds.

2. Design Rationale

- **3rd-Order Passive Low-Pass Filter**:
    - The top network (R1-R3, C1-C3 for the positive rail and R6-R8, C4-C6 for the negative) forms a **differential 3rd-order RC low-pass filter**.
    - This is essential for PDM signals, as it aggressively suppresses the high-frequency quantization noise (carrier frequency) and recovers the baseband analog audio envelope. A 3rd-order slope provides a much sharper cutoff than simpler filters, resulting in a cleaner signal for the amplifier.
- **Amplifier Configuration (NS4150B)**:
    - The **NS4150B** is a high-efficiency Class-D amplifier chosen for its low EMI and differential input/output capabilities, which are ideal for the noisy environment of a mobile robot.
    - **Gain Setting**: The gain is calculated as $Gain = 240 / 51 = 7.2$, a value chosen to maximize volume without clipping, based on the speaker's impedance and the PDM signal's voltage swing.
- **Pop Noise Suppression (PA_CTL)**:
    - A **10kΩ pull-down resistor (R11)** is connected to the **CTRL (Pin 1)**. This ensures the PA remains in a low-power shutdown mode during the MCU's boot-up phase, preventing the "pop" sounds often caused by transient GPIO states.
- **AC Coupling and Bias**:
    - **C13 and C15 (100nF)** are used as **AC coupling capacitors** to block any DC offset from the filter stage.
    - **C14 (1uF)** on the **Bypass (Pin 2)** provides a stable internal reference voltage, which is critical for reducing Total Harmonic Distortion (THD).
- **Power Filtering**: **C16 (100nF)** and **C17 (1uF)** are placed immediately adjacent to the **VCC (Pin 6)** to handle high-frequency transients and provide bulk decoupling, ensuring the amplifier remains stable during high-output peaks.

3. Expected Outcome

The implementation of this differential audio path results in a robust audio system that is highly resistant to common-mode interference from the robot's motors and Wi-Fi radio. The 3rd-order filtering ensures a smooth, natural sound quality suitable for human-robot interaction, while the hardware-level shutdown control (PA_CTL) ensures a professional user experience by eliminating unwanted noise during power-on cycles.



LCD Interface and Visual Feedback System

1. Functional Overview

This schematic section defines the interface for the **ST7789 LCD**, which serves as the robot's primary visual output for displaying facial expressions and system status. The connection is established via a **10-pin FPC (Flexible Printed Circuit)** connector, utilizing a high-speed serial interface to ensure smooth animation and low-latency interaction.

2. Design Rationale

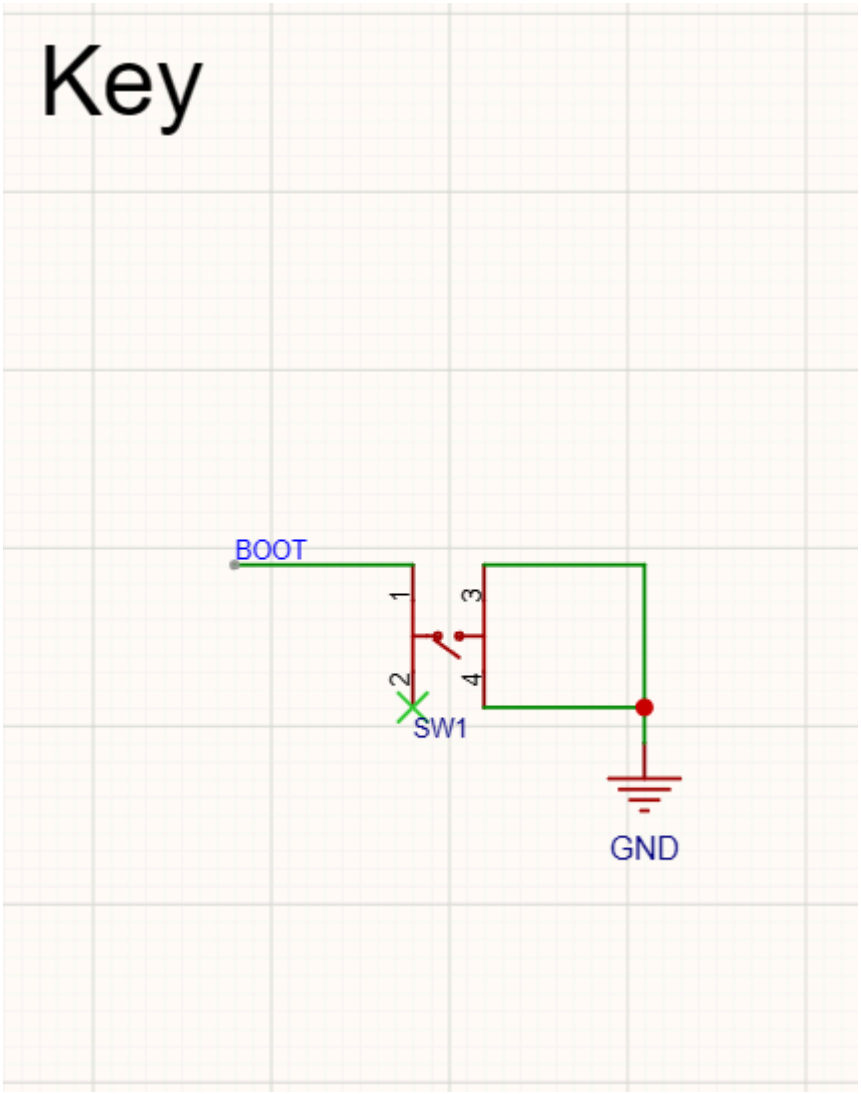
- **Interface Selection**: The display uses a 4-wire serial interface (SDA, SCL, RS/DC, CS). This configuration was chosen to minimize the number of GPIO pins required on the ESP32-S3 while maintaining a high enough refresh rate for fluid "desk fatigue" combat animations.
- **Backlight Current Limiting**:
    - **R15 (10Ω)** is a crucial current-limiting resistor for the **LCD_LEDA** (Anode) pin.

    - It ensures the backlight LEDs operate within their safe current range, preventing thermal degradation and extending the lifespan of the display panel.
- **Control Logic & Stability**:
    - **R16 (10kΩ)** acts as a pull-up for the **LED_RST** (Reset) line, ensuring the display remains in a stable state and does not inadvertently reset due to floating signal noise.
    - **R17 (10kΩ)** is used as a pull-down for **LED_CS** (Chip Select) to keep the peripheral deselected by default, preventing bus contention during the MCU's boot-up phase.
- **Power Decoupling**: **C29 (100nF)** is placed immediately adjacent to the **MCU_3V3** input of the FPC connector. This filters out high-frequency switching noise from the digital data lines, ensuring a clean power supply for the display's internal driver IC.

3. Expected Outcome

The integration of this LCD module allows the robot to convey "emotions" through various facial expressions, a core requirement for its role in reducing user fatigue. The hardware-level current limiting and pull-up/down resistors ensure that the display is both electrically robust and reliable over long periods of operation. By utilizing a compact FPC connector, the design maintains a small physical footprint, fitting perfectly within the robot's "face board" architecture.

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Manual Input and Boot Control Interface

1. Functional Overview

This schematic section highlights the **Manual Tactile Switch (SW1)**, which is configured as the **BOOT button** for the robot. Its primary role is to provide a physical interface for the user to interact with the ESP32-S3's boot sequence, enabling manual firmware flashing or triggering user-defined software interruptions during the robot's operation.

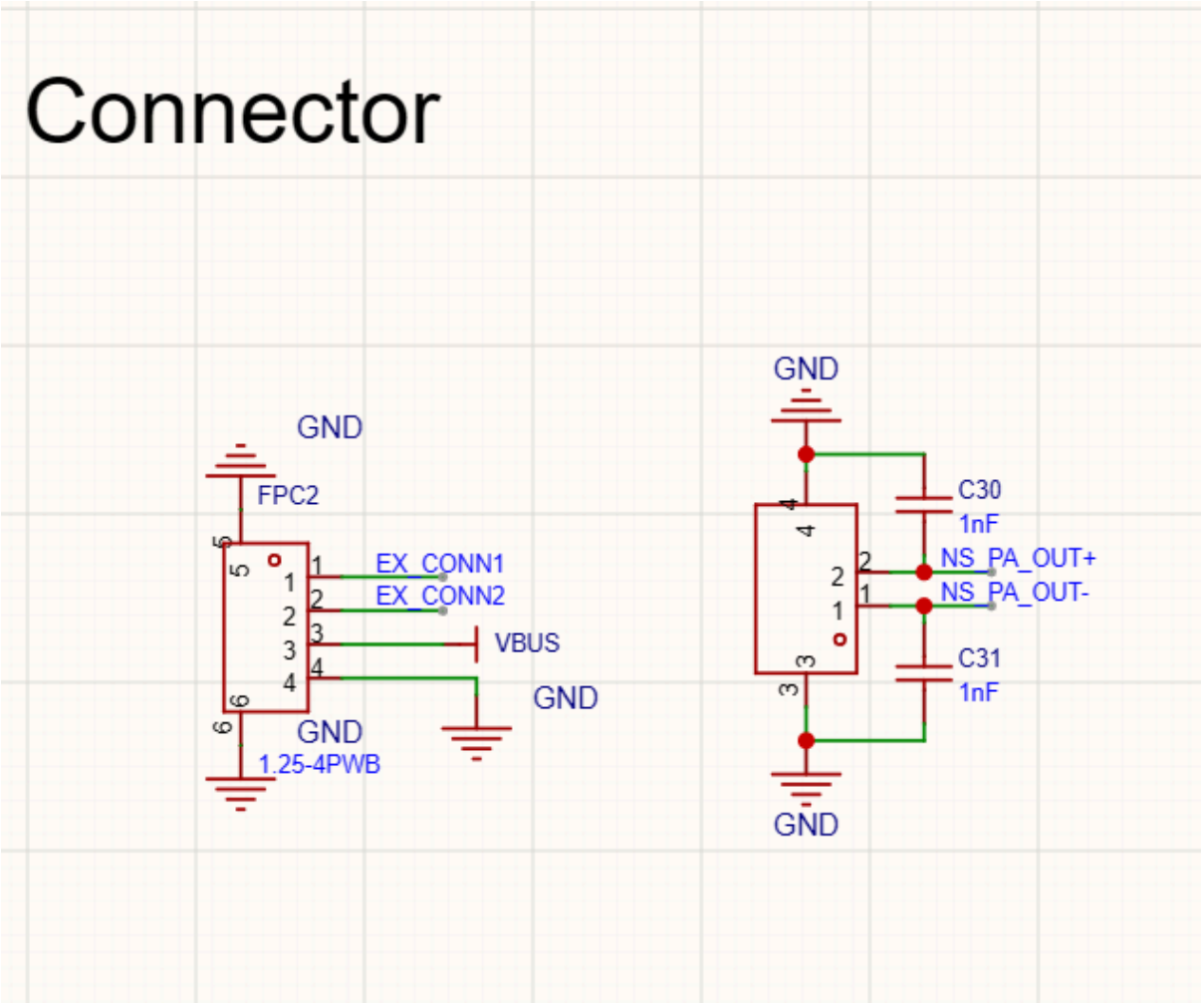
2. Design Rationale

- **Active-Low Logic**: The switch is designed in an **active-low configuration**. One side is tied to the **BOOT (IO0)** signal of the ESP32-S3, while the other side is connected directly to **GND**. When the button is pressed, it pulls the IO0 pin to a logic-low level.
- **Bootstrap Functionality**: Per the ESP32-S3 hardware specifications, holding the BOOT pin low during power-up or reset forces the SoC into **Serial Bootloader Mode**. This hardware-level entry point is essential for initial firmware deployment and recovery if the application code becomes corrupted.

- **GPIO Reusability**: Beyond its role during the boot sequence, this switch is mapped to **IO0**, allowing it to be used as a general-purpose functional button (e.g., "Mode Switch" or "Emergency Stop") once the robot has successfully booted into its main application.
- **Mechanical Integrity**: The use of a standard 4-pin tactile switch footprint provides superior mechanical stability on the PCB compared to 2-pin variants, ensuring the component can withstand repeated manual presses during the development and testing phases.

3. Expected Outcome

The inclusion of this manual switch ensures that the robot is always accessible for programming, even when wireless (OTA) updates are not feasible. It provides a reliable fail-safe for the developer and adds a layer of manual control for the end-user. By grounding the signal directly upon activation, the design guarantees a clean logic transition, enabling the ESP32-S3 to accurately detect user input for state changes or bootloader entry.



Peripheral Connectors and Signal Interfacing

1. Functional Overview

This schematic section details the **External Connector (FPC2)** and the **Speaker Output Terminal**. These components serve as the physical interface between the main PCB and external hardware modules. The design facilitates power distribution and high-fidelity audio transmission while maintaining modularity in the robot's hardware architecture.

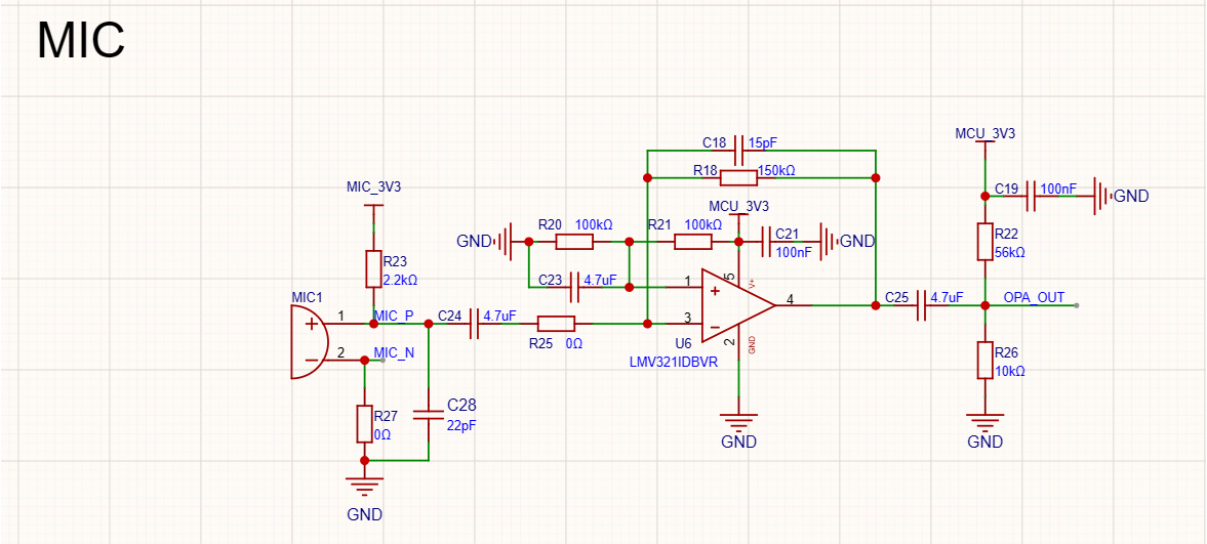
2. Design Rationale

- **Expansion Port (FPC2/1.25-4PWB)**:
    - This 4-pin connector provides access to **VBUS (5V)** and **GND**, along with two general-purpose signal lines (**EX_CONN1**, **EX_CONN2**).
    - The choice of a **1.25mm pitch connector** (or FPC equivalent) balances spatial efficiency with the mechanical robustness required for connecting sub-boards, such as the robot's "body board" or specialized sensor arrays.
- **Speaker Output & EMI Filtering**:
    - The output terminal carries the amplified differential audio signals (**NS_PA_OUT+** and **NS_PA_OUT-**) directly from the NS4150B power amplifier.

    - **C30 and C31 (1nF)** are implemented as **EMI suppression capacitors**. By shunting high-frequency switching noise from the Class-D amplifier to **GND**, these capacitors prevent the speaker wires from acting as antennas, thereby reducing electromagnetic interference (EMI) that could affect the sensitive I2S microphone or Wi-Fi signals.
- **Common Grounding**: Both connectors share a unified **GND** plane to ensure a stable reference voltage and minimize ground loops between interconnected modules.

3. Expected Outcome

This interfacing strategy ensures the robot remains modular and easy to assemble. The inclusion of dedicated filtering on the audio output ensures high-quality sound reproduction without compromising the integrity of other high-speed digital signals on the board. By providing a standardized expansion port, the design allows for future hardware upgrades or additional sensor integration with minimal redesign.



Microphone Preamplifier and Analog Signal Conditioning

1. Functional Overview

This schematic section details the **Analog Microphone Preamplifier** circuit. Its primary role is to capture low-amplitude acoustic signals from the electret microphone (MIC1) and amplify them to a level suitable for the ESP32-S3's Analog-to-Digital Converter (ADC). This module is essential for the robot's voice interaction capabilities, providing the raw audio data needed for **ASR (Automatic Speech Recognition)** and **Cloud AI** processing.

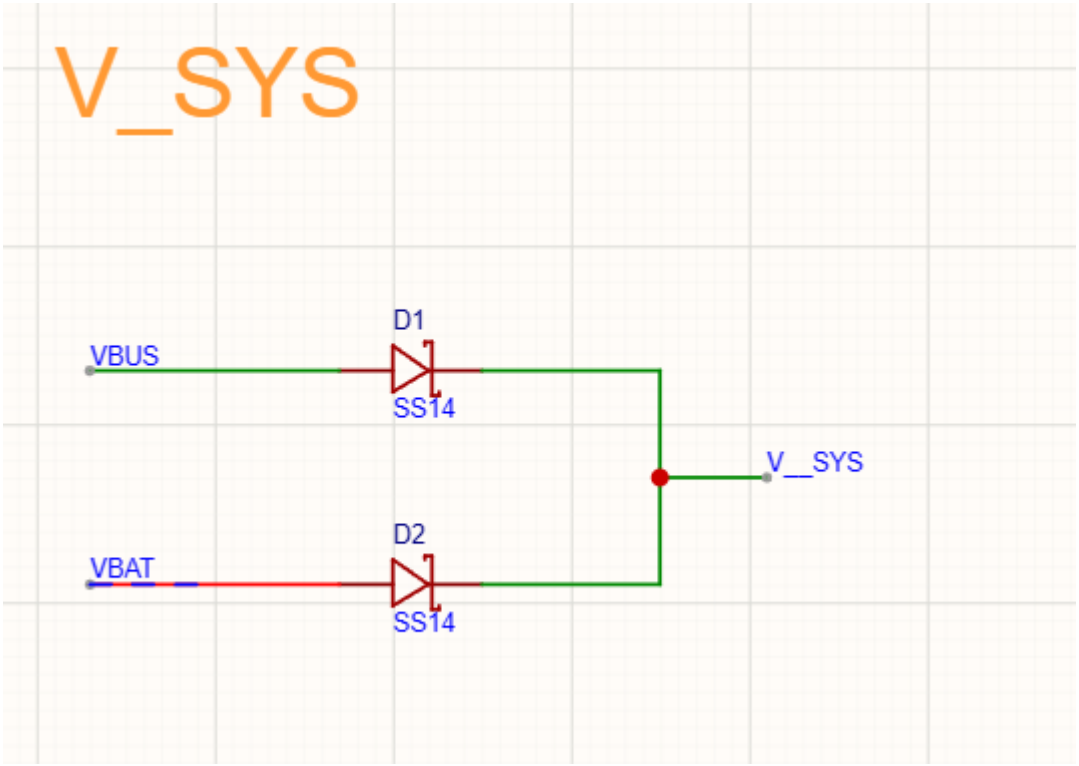
2. Design Rationale

- **Microphone Biasing**:
  - **R23 (2.2kΩ)** provides the necessary DC bias current from the filtered **MIC_3V3** rail to the electret microphone.
  - Using the dedicated **MIC_3V3** supply (which, as noted in the LDO section, is isolated via an RC filter) is critical here to prevent digital switching noise from appearing as audible hiss in the recordings.
- **Non-Inverting Amplification (LMV321)**:
  - The **LMV321IDBVR** is a rail-to-rail operational amplifier chosen for its low power consumption and small footprint.
  - **Gain Calculation**: The gain is determined by the feedback network. With **R18 (150kΩ)** and **R21 (100kΩ)** in a non-inverting configuration, the gain is approximately $1 + (150/100) = 2.5$.
  - **C18 (15pF)** in parallel with the feedback resistor acts as a high-frequency pole to prevent oscillations and filter out electromagnetic interference (EMI).
- **DC Offsetting and Biasing**:
  - Since the ESP32-S3 ADC can only measure positive voltages, the audio signal must be "lifted" to a mid-rail reference.
  - **R20 and R21 (100kΩ each)** form a voltage divider that sets the common-mode voltage to **1.65V** ($VCC/2$).
  - **C23 (4.7uF)** provides a low-impedance AC ground for this bias point, ensuring the reference remains stable during signal swings.
- **AC Coupling**:

    - **C24 (4.7uF)** and **C25 (4.7uF)** are used as input and output coupling capacitors, respectively. They block DC offsets while allowing the AC audio signal to pass through the amplification stages.
- **Output Protection and Filtering**:
    - **R22 and R26** form a final voltage divider/protection stage at **OPA_OUT** to ensure the signal remains within the safe input range of the MCU's ADC pins.
    - **C19 (100nF)** provides final high-frequency decoupling for the amplifier's power supply pin.

3. Expected Outcome

The resulting preamplifier circuit delivers a centered, high-fidelity analog signal with minimal noise floor. By leveraging the isolated power rail and carefully calculated gain, the system can reliably capture human speech even in environments with moderate ambient noise. This hardware-level conditioning significantly improves the accuracy of the robot's subsequent digital signal processing and AI-driven voice responses.



Power Path Management and Voltage ORing

1. Functional Overview

This schematic section illustrates the **Power Path Management** circuit, which utilizes a dual-diode ORing configuration to create the main system power rail, **V_SYS**. It allows the robot to seamlessly switch between two power sources—**VBUS** (USB 5V) and **VBAT** (Battery)—ensuring that the system remains powered as long as at least one source is available.

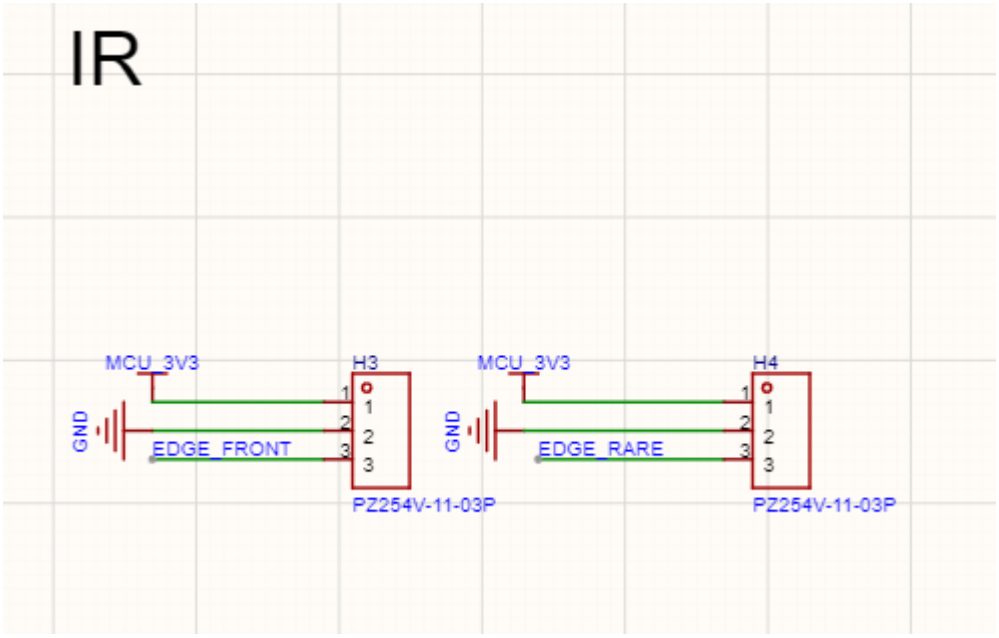
2. Design Rationale

- **Schottky Diode Selection (SS14)**:
  - Two **SS14 Schottky diodes** (**D1** and **D2**) are used for their low forward voltage drop (V_f) and fast switching speeds.
  - In a battery-powered robotics application, minimizing the voltage drop across the diodes is critical to maximize energy efficiency and ensure sufficient headroom for the downstream 3.3V LDO regulator.
- **Reverse Current Protection**:
  - The diodes act as one-way valves. This prevents **VBUS** from inadvertently charging the battery in an uncontrolled manner (which could be hazardous) and prevents the battery from back-powering the USB port of a connected computer.
- **Automatic Source Prioritization**:
  - The circuit naturally implements a "highest voltage wins" logic. If both USB and a fully charged battery are connected, the source with the higher potential will supply the load, effectively reducing the drain on the battery when a tethered power source is available.
- **Simplicity and Reliability**:

- This passive ORing approach was chosen over complex Power Path ICs to reduce the Bill of Materials (BOM) count and PCB complexity while providing a robust, fail-safe mechanism for the robot's basic power needs.

3. Expected Outcome

This design ensures the robot's "brain" and peripherals have a continuous power supply during transitions, such as unplugging the USB cable to let the robot move autonomously. The use of SS14 diodes provides a balance between cost-effectiveness and electrical performance, protecting the delicate internal electronics from reverse-polarity damage and source contention.



Infrared (IR) Sensor Interface for Edge Detection

1. Functional Overview

This schematic section defines the interface for the robot's **Infrared (IR) sensors** via two 3-pin connectors (**H3** and **H4**). These sensors are strategically positioned to serve as an **edge detection system** (cliff sensors), providing critical environmental feedback to the ESP32-S3 main controller to prevent the robot from falling off surfaces such as desks or tables.

2. Design Rationale

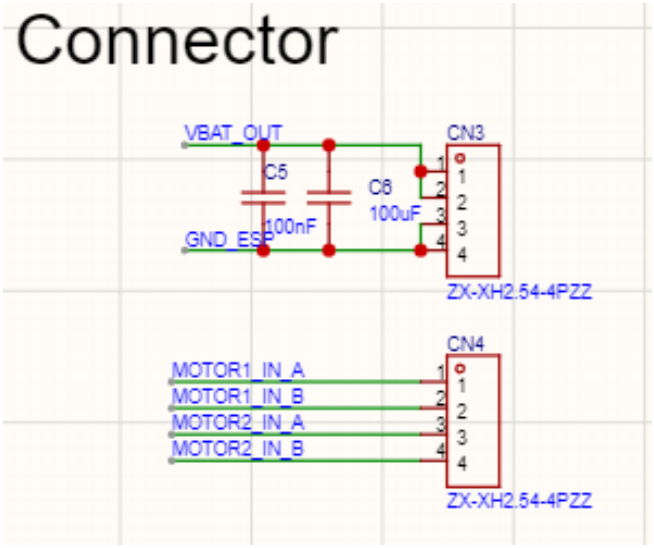
- **Sensor Selection (TCRT5000)**:
  - The design utilizes **TCRT5000 IR reflective sensors**.
  - These were selected to replace the initial Time-of-Flight (ToF) sensor concept to provide a more cost-effective and reliable solution for short-range surface detection.
- **Interface Configuration**:
  - Each connector utilizes a standard 3-pin layout: **Pin 1 (MCU_3V3)** for power, **Pin 2 (GND)** for the common ground, and **Pin 3** for the signal return (**EDGE_FRONT** and **EDGE_RARE**).
  - This standardized wiring allows for easy maintenance and ensures that the sensors can be mounted on the "body board" while communicating directly with the main controller.
- **Signal Mapping**:
  - The **EDGE_FRONT** and **EDGE_RARE** signals are routed to the ESP32-S3's GPIOs (**IO17** and **IO18**).
  - This enables the firmware to monitor the surface status in real-time at both the leading and trailing edges of the robot's chassis.
- **Power Stability**:
  - The sensors are powered by the **MCU_3V3** rail.

- As established in the LDO design section, this rail is heavily decoupled to ensure that the IR emitters and receivers do not introduce switching noise into the more sensitive analog components, such as the microphone preamplifier.

3. Expected Outcome

The implementation of these IR interfaces creates a robust "Safety Priority Layer" for the robot's motion control. By detecting the absence of a reflective surface (indicating a desk edge), the ESP32-S3 can execute an immediate "stop and reverse" command, bypassing other UI or manual controls to ensure the physical integrity of the hardware. This hardware-level safety feature is fundamental for a desktop robot designed to operate autonomously in potentially hazardous table-top environments.

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Week 2: Motor Power and PWM Control Interconnect

1. Functional Overview

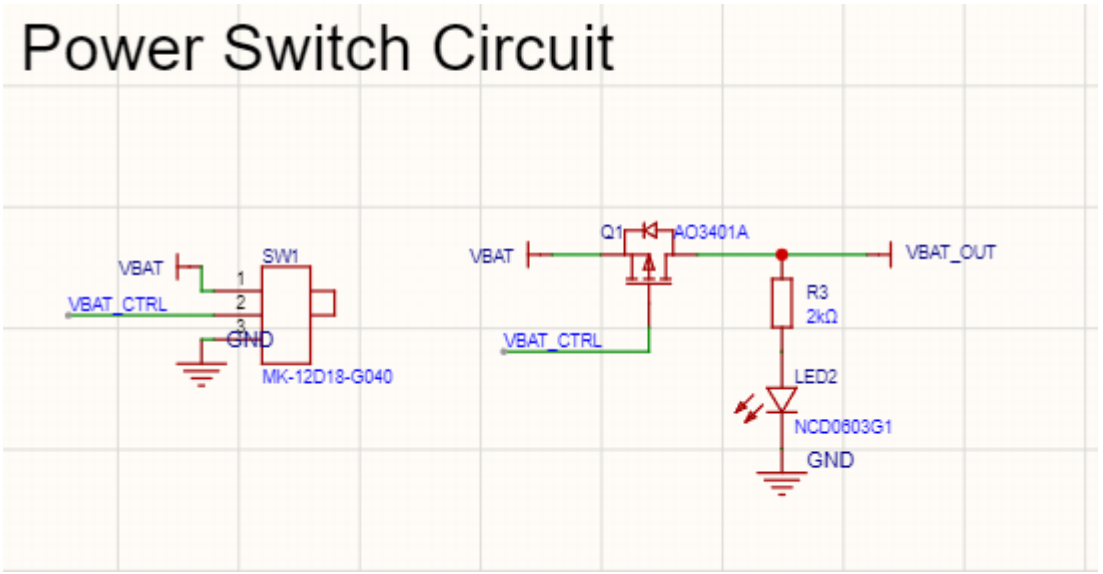
This schematic section highlights the **Motor Power and Signal Connectors (CN3, CN4)**, which form the critical bridge for high-current power and PWM control logic. By separating the power supply and logic signals into distinct connector blocks, the design minimizes cross-talk and ensures that the large currents required by the motors do not interfere with the sensitive PWM timing signals sent from the ESP32-S3.

2. Design Rationale

- **Dedicated Motor Power Rail (CN3)**:
    - **CN3** is dedicated to the **VBAT_OUT** supply, providing the primary energy source for the motors.
    - **Bulk Decoupling**: The inclusion of **C6 (100uF)** and **C5 (100nF)** is essential for a dual-PCB robotics system. The 100uF electrolytic capacitor acts as a local energy reservoir to handle the high inrush current when the motors start or reverse, preventing a sudden voltage drop that could reset the MCU. The 100nF capacitor filters out high-frequency noise generated by the motor's internal brushes.
- **PWM Logic Interface (CN4)**:
    - **CN4** carries the four logic signals (**MOTOR1_IN_A/B**, **MOTOR2_IN_A/B**) required to drive the TC118S H-bridge drivers.
    - These signals are standard **Pulse Width Modulation (PWM)** outputs from the ESP32-S3, allowing for precise speed control and directional switching (Forward, Backward, Brake) for the robot's differential drive system.
- **Physical Robustness**: The use of **XH2.54-4PZZ** connectors provides a secure, vibration-resistant latching mechanism. This is vital for internal robot wiring, as it ensures electrical continuity even during the mechanical stress of rapid movement or "desk fatigue" combat maneuvers.

3. Expected Outcome

This dual-connector architecture provides a clean electrical separation between high-power and high-speed logic circuits. The localized filtering on the power rail ensures system stability, while the dedicated PWM signal path guarantees that the motor drivers receive clear, distortion-free commands. This setup is the foundation for the robot's responsive and agile movement profile, enabling smooth acceleration and accurate positioning.



Week 2: Power Switch Circuit and Battery Load Management

1. Functional Overview

This schematic details the **Power Switch Circuit**, which acts as the primary hardware interlock for the robot's battery power. It combines a physical sliding switch (**SW1**) with a high-side P-Channel MOSFET (**Q1**) to control the flow of energy from the battery (**VBAT**) to the rest of the system (**VBAT_OUT**). This setup provides a reliable way to completely de-energize the robot's high-power components, such as the motors, while preventing unnecessary battery drain when the device is not in use.

2. Design Rationale

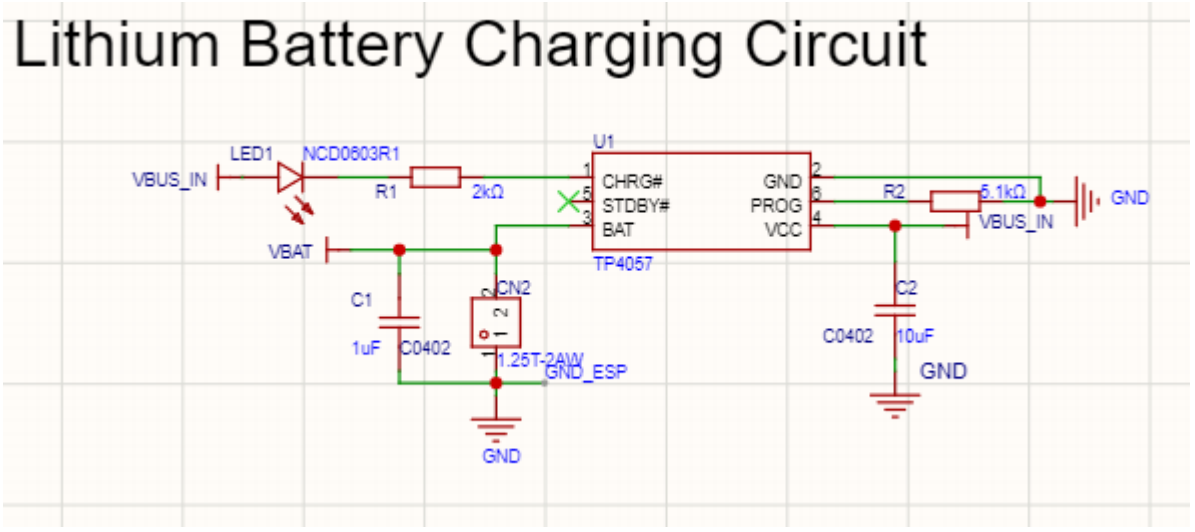
- **High-Side P-Channel MOSFET Switch (Q1)**:
    - The **AO3401A** P-Channel MOSFET is the core switching element. It was selected for its low $R_{DS(ON)}$, which minimizes power loss and heat generation when the robot is drawing high current for motor maneuvers.
    - By using a MOSFET to handle the actual load current, the physical switch (**SW1**) only needs to handle a very small signal-level current to toggle the

MOSFET's gate. This prevents the mechanical contacts of the switch from degrading due to the high inrush currents typical of DC motors.

- **Control Logic (SW1)**:
  - **SW1 (MK-12D18-G040)** is a slide switch that controls the **VBAT_CTRL** line.
  - When the switch connects **VBAT_CTRL** to **GND**, the gate of **Q1** is pulled low, turning the MOSFET "ON" and allowing power to flow to **VBAT_OUT**.
  - When switched to position 1, **VBAT_CTRL** is tied to **VBAT**, turning the MOSFET "OFF."
- **Visual Status Indicator (LED2)**:
  - The **NCD0603G1** LED, paired with a **2kΩ current-limiting resistor (R3)**, is connected directly to the **VBAT_OUT** rail.
  - This provides an immediate visual confirmation that the system is energized, serving as a critical safety feature for the user.
- **Safety and Protection**:
  - The circuit utilizes a common **GND** for all control signals to ensure logic stability.
  - The use of the AO3401A provides a degree of inherent protection against transient spikes from the motor rails during power-down cycles.

3. Expected Outcome

This power management strategy ensures that the robot can be safely turned on and off without risking mechanical wear on the primary control switch. The low resistance of the AO3401A ensures that the full potential of the battery is delivered to the **TC118S motor drivers**, enabling maximum torque and efficiency. Additionally, the integrated LED indicator provides a professional "power-on" feedback mechanism, improving the overall user experience and safety of the desktop robot.



Week 2: Lithium Battery Charging Circuit

1. Functional Overview

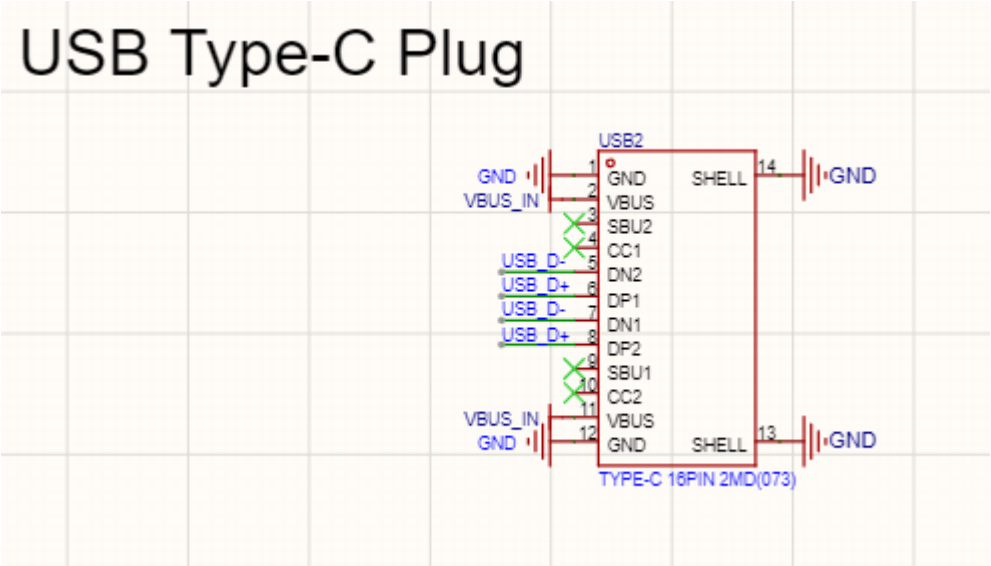
This schematic illustrates the **Lithium Battery Charging Circuit**, centered around the **TP4057** linear charger IC. This subsystem manages the safe replenishment of the robot's onboard lithium battery using power provided by the **VBUS_IN** (USB 5V) rail. It handles the entire charging profile—from trickle charge to constant current and constant voltage stages—ensuring the battery remains healthy and ready for autonomous "desk fatigue" combat maneuvers.

2. Design Rationale

- **Charging IC Selection (TP4057)**:
    - The **TP4057** was chosen for its simplicity, small SOT-23-6 footprint, and integrated thermal feedback, which prevents the IC from overheating during high-current charging cycles.
    - It requires no external sense resistor or blocking diode, saving valuable PCB real estate.
- **Charge Current Programming**:
    - The charging current is set by the **PROG (Pin 6)** resistor **R2 (5.1kΩ)**.
    - Based on the TP4057 formula ($I_{bat} = (V_{prog} / R_{prog}) \times 1000$), a 5.1kΩ resistor sets a safe and efficient charging rate of approximately **200mA**, suitable for the target battery capacity while minimizing heat.
- **Visual Status Indicator (LED1)**:
    - The **NCD0603R1** (Red LED) and **R1 (2kΩ)** are connected to the **CHRG# (Pin 1)** open-drain output.
    - This LED illuminates while the battery is actively charging and extinguishes once the charge cycle is complete, providing clear user feedback.
- **Power and Stability**:
    - **C2 (10uF)** and **C1 (1uF)** provide decoupling for the **VCC** and **BAT** pins, respectively. These capacitors suppress voltage ripples and ensure the charging control loop remains stable regardless of transients on the USB power line.
- **Battery Interconnect (CN2)**:
    - A **1.25T-2AW** connector is used for the physical battery connection (**VBAT** and **GND**). This ensures a secure, polarized attachment that prevents accidental reverse-polarity damage during assembly.

3. Expected Outcome

This charging circuit provides a reliable and autonomous power replenishment system for the robot. By utilizing the TP4057, the design guarantees that the lithium battery is never overcharged, significantly extending its operational lifespan. The integrated LED and standardized connector make the system user-friendly and easy to maintain, while the optimized 200mA charge rate ensures the robot can be fully recharged from a standard USB port in a reasonable timeframe without excessive thermal stress.



Week 2: USB Type-C Plug for Dedicated Charging Input

1. Functional Overview

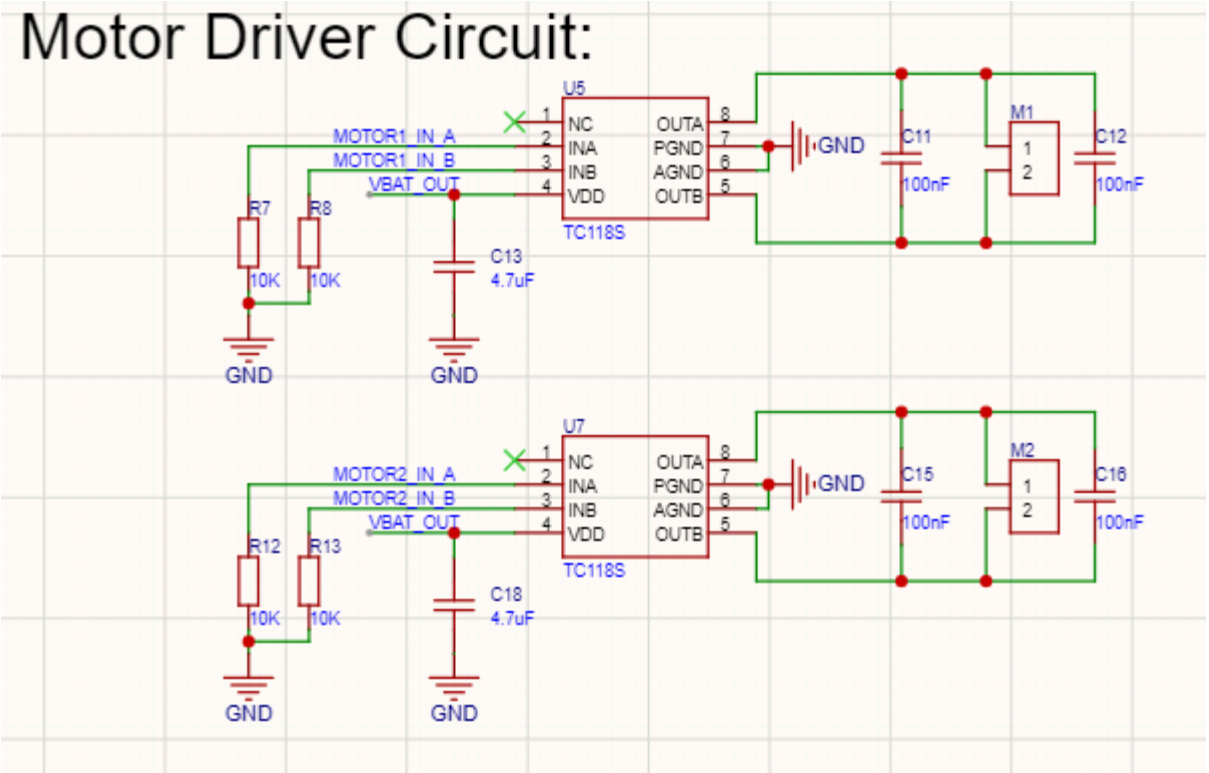
This schematic section features a **USB Type-C Plug (16-pin)**, which serves as the robot's dedicated **power input port**. Specifically designed for charging the onboard lithium battery, this interface accepts 5V from an external USB power source via the **VBUS_IN** lines. By utilizing the modern USB-C form factor, the robot benefits from a robust and reversible connection point for maintaining its energy reserves.

2. Design Rationale

- **Charging-Centric Design**: While the connector includes data pins, its primary implementation in this system is to provide a reliable path for **VBUS_IN**. This 5V supply is routed directly to the **TP4057 charging circuit** to replenish the battery, which then powers the high-current components like the **TC118S motor drivers**.
- **Power Handling and Efficiency**:
    - Pins 2 and 11 are tied to **VBUS_IN**. Doubling these pins allows the connector to handle charging currents with lower resistance, reducing heat generation during the charge cycle.
    - The symmetric layout ensures that the robot can be charged regardless of the cable's orientation, enhancing user convenience.
- **Mechanical Durability**: The **SHELL** pins (13, 14) are securely grounded. This provides critical mechanical reinforcement for the connector, ensuring it remains firmly attached to the PCB even after repeated insertion and removal of the charging cable.
- **Signal Isolation**: By isolating the charging input through this dedicated plug, the design ensures that high-power charging transients do not interfere with the internal communication or the PWM logic signals used to control the robot's movement.

3. Expected Outcome

The implementation of this USB-C plug ensures that the robot has a standardized and efficient method for battery charging. By dedicating this interface to power delivery, the design maintains a clear separation between charging operations and internal system logic. This setup guarantees that the robot remains ready for action, providing the necessary energy for the motors to combat "desk fatigue" while offering a professional, durable charging experience for the user.



Week 2: Dual-Channel Motor Driver System

1. Functional Overview

This schematic section highlights the **Dual Motor Driver Circuit**, utilizing two **TC118S** H-bridge driver ICs to control the robot's locomotion. These drivers translate low-power **PWM signals** from the ESP32-S3 into high-current outputs required to drive the DC gear motors (M1 and M2). This configuration enables differential steering, allowing the robot to move forward, backward, and execute precise turns to interact with the user.

2. Design Rationale

- **H-Bridge Driver (TC118S)**:
    - The **TC118S** is a single-channel integrated H-bridge driver designed for high efficiency and low voltage drops.

    - By using two independent ICs (**U5 and U7**), the design achieves full independent control over both wheels, which is essential for the robot's agility.
- **Logic Protection and Stability**:
    - **Pull-down Resistors (R7, R8, R12, R13)**: $10\text{k}\Omega$ resistors are connected to the **INA** and **INB** inputs of each driver. These ensure the motors remain in a "High-Z" or coast state if the MCU pins are floating during boot-up, preventing unintended robot movement.
    - **Input Filtering**: **C13 and C18 ($4.7\mu\text{F}$)** capacitors are placed near the **VDD** pins to stabilize the **VBAT_OUT** supply, protecting the drivers from voltage dips caused by sudden current draws.
- **Motor Noise Suppression**:
    - **C11, C12, C15, and C16 (100nF)** are implemented across the motor terminals and to ground.
    - These capacitors filter out high-frequency electromagnetic interference (EMI) and brush noise generated by the DC motors. This is critical to prevent electrical noise from coupling into the sensitive **I2S microphone** or disrupting the **LCD** data signals.
- **Power Efficiency**: The drivers are powered by **VBAT_OUT**, which is the switched battery rail managed by the P-Channel MOSFET. This ensures that when the main power switch is off, the motor drivers are completely de-energized, preserving battery life.

3. Expected Outcome

The implementation of this dual-driver architecture provides a robust and responsive drivetrain for the desktop robot. The combination of hardware pull-downs and EMI filtering ensures that the robot's movement is predictable and does not interfere with its "face board" electronics. This system allows for the high-torque maneuvers necessary to effectively "combat" desk fatigue through expressive and dynamic physical motion.

week 3.30 syc

Weekly Progress Log – Week 3/30

Project: Interactive Desktop Companion Robot
Team: Jiajun Gao, Yuchen Shih, Zichao Wang

This Week's Progress

This week, we focused on the mechanical design and overall physical layout of the robot. Based on the dimensions of the PCB that arrived, we measured the board carefully and used it as the foundation for planning the robot's external structure and internal component arrangement.

1. PCB-Based Dimension Measurement and Layout Planning

After receiving the PCB, we measured its physical dimensions and used those measurements to guide the robot's structural design. This step was important because the enclosure and internal layout must match the actual PCB size to ensure that all electrical and mechanical components can fit properly within the robot body.

Using the PCB as a reference, we began planning the spatial arrangement of the robot's internal system, including how the board would be housed and how other components would be positioned relative to it.

2. Robot Exterior Design in Fusion 360

We designed the overall appearance of the robot using **Fusion 360**. The design focused on the two main structural parts of the robot: the **head** and the **body**. At this stage, the goal was to establish the robot's general form while also ensuring that the design could realistically accommodate the internal hardware.

The CAD design process allowed us to begin translating the functional prototype into a more complete physical product form.

3. Reserved Space for Key Hardware Components

During the design process, we also reserved space for major hardware components to ensure the feasibility of later assembly. Specifically, we included designated positions for:

- the **wheels**
- the **motors**
- the **front-facing sensor**
- the **downward-facing sensor**

These reserved locations are important for integrating both the mobility and safety functions of the robot into the final physical design. The front-facing sensor is used for obstacle

detection, while the downward-facing sensor monitors the distance between the robot and the desk surface to support desk-edge detection and prevent the robot from falling.

Key Achievements

- Measured the actual PCB dimensions after arrival
- Used PCB dimensions to guide structural and spatial design decisions
- Created the initial robot exterior design in Fusion 360
- Designed the head and body structure of the robot
- Reserved mounting space for wheels, motors, the front-facing sensor, and the downward-facing sensor

Challenges

- The internal layout must balance compact size with enough space for all hardware components
- Component placement needs to support both functional performance and mechanical stability
- The sensor locations need to be carefully aligned with the robot's obstacle detection and desk-edge detection requirements
- Additional refinement is needed to ensure the enclosure design matches real assembly constraints

Next Steps

- Refine the Fusion 360 model for better structural integration
- Confirm the detailed placement and mounting method of each component
- Continue adjusting the enclosure design based on mechanical and electrical requirements
- Prepare for the next stage of physical assembly and hardware integration

week 4.6 syc

Weekly Progress Log – Week 4/6

Project: Interactive Desktop Companion Robot
Team: Jiajun Gao, Yuchen Shih, Zichao Wang

This Week's Progress

This week, we focused on the initial hardware fitting, physical assembly validation, and iterative refinement of the robot structure. After completing the basic exterior design in Fusion 360 last week, we began testing whether the major internal components could be properly placed inside the designed robot structure. Based on the fitting results, we modified multiple design versions to make the dimensions and overall operating logic as compatible as possible.

1. Initial Motor and PCB Placement Test

We attempted to place the motors and PCB into the designed robot body to check whether the internal layout was feasible. This step allowed us to evaluate whether the reserved spaces from the previous design stage were accurate enough for real hardware integration.

By physically testing the PCB and motor placement, we were able to better understand the assembly constraints, including component clearance, wire routing space, and the overall fit between the electronic components and the robot enclosure.

2. Head Design Refinement for Interaction Modules

We also refined the robot head design by adding openings for several interaction-related components. Specifically, we designed openings for:

- the **screen**
- the **microphone**
- the **speaker**

These openings are important because the head of the robot is intended to support future interaction functions, such as visual display, sound input, and audio output. Adding these openings at this stage helped us prepare the mechanical structure for later integration of the robot's interactive features.

3. Trial Assembly and Fit Checking

After placing the PCB, motors, and interaction-related openings into the design, we attempted an initial trial assembly to check whether the overall structure was reasonable. The purpose of this step was to see whether the components could fit properly inside the robot body and head, and whether the dimensions were accurate enough for physical assembly.

This fitting process helped us identify whether the robot enclosure was too tight, too loose, or needed further adjustment. It also provided useful feedback for improving the CAD model before moving toward a more complete assembly.

4. Iterative Design Modification

Throughout this week, we created and modified multiple versions of the robot structure. These changes were made to improve the match between the physical dimensions of the components and the robot's intended operating logic.

For example, the body design had to provide enough space for the PCB, motors, and wiring, while the head design needed to support the screen, microphone, and speaker openings without affecting the overall appearance. By comparing different design versions, we were able to gradually improve the fit between the mechanical structure and the internal hardware layout.

Key Achievements

- Tested the initial placement of the PCB inside the robot body
- Checked whether the motors could fit into the reserved motor positions
- Refined the head design by adding openings for the screen, microphone, and speaker
- Performed a trial assembly to evaluate the overall physical fit
- Modified multiple design versions to improve dimensional accuracy and hardware compatibility
- Improved the connection between the mechanical structure and the robot's intended operating logic

Challenges

- The PCB and motors must fit inside a compact enclosure while still leaving enough space for wiring
- The screen, microphone, and speaker openings need to be accurately positioned for both function and appearance
- Some component spaces required multiple rounds of adjustment to improve assembly tolerance
- The overall structure needs to balance internal hardware requirements, external appearance, and functional logic

Next Steps

- Continue refining the CAD model based on the trial assembly results
- Adjust component openings and mounting positions if necessary
- Improve internal spacing for wires and hardware connections
- Continue testing the fit between the robot enclosure and the actual electronic components
- Prepare for a more complete physical assembly of the robot

week 4.13 syc

Weekly Progress Log – Week 4/13

Project: Interactive Desktop Companion Robot
Team: Jiajun Gao, Yuchen Shih, Zichao Wang

This Week's Progress

This week, we focused on validating the final physical fit of the 3D-printed robot enclosure and testing the integration of all major hardware modules. After several rounds of design refinement in previous weeks, the overall dimensions and 3D-printed exterior successfully met our design requirements. The final enclosure size was very close to the intended measurements, and all modules were able to fit into their corresponding positions smoothly.

1. Final 3D-Printed Enclosure Validation

We tested the 3D-printed robot exterior and confirmed that the overall size, shape, and internal spacing were suitable for the robot's hardware layout. The printed structure matched our design expectations well, and the dimensions were accurate enough for real assembly.

This was an important milestone because it confirmed that our Fusion 360 design could be translated into a physical enclosure that supports the actual hardware components. The head and body structures both provided enough space for their assigned modules while maintaining the intended overall appearance of the robot.

2. Complete Hardware Module Placement

We successfully placed all major hardware modules into their designed positions inside the robot. These components included:

- the **PCB in the head**
- the **PCB in the body**
- the **battery in the head**
- the **battery in the body**
- the **front-facing sensor**
- the **downward-facing sensor**
- the **screen in the head**
- the **microphone**
- the **speaker**

All components were able to fit into their corresponding locations as planned. This confirmed that the mechanical design, internal layout, and component placement were well aligned with the actual hardware dimensions.

3. Power Supply and Electrical Integrity Check

After placing the modules into the enclosure, we powered on the system to verify that all major modules could receive power correctly. This step helped us confirm the basic electrical functionality of the integrated system.

We also checked the soldering quality and overall circuit integrity to make sure that the electrical connections were complete and stable. This included verifying whether the main boards, sensors, screen, microphone, speaker, and batteries were properly connected and able to support the next stage of testing.

Key Achievements

- Verified that the 3D-printed robot exterior met the design requirements
- Confirmed that the overall dimensions were accurate and suitable for assembly
- Successfully placed all major modules into their corresponding positions

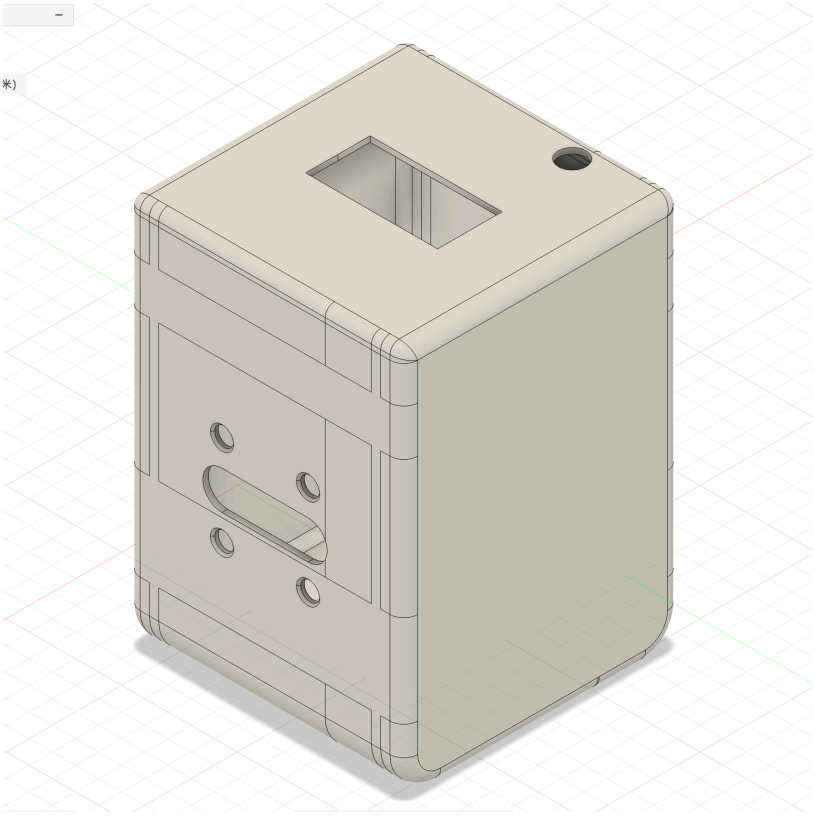
- Integrated the head PCB, body PCB, head battery, body battery, sensors, screen, microphone, and speaker into the enclosure
- Powered on the system and confirmed that the modules could receive power
- Checked the soldering and overall circuit integrity of the robot system

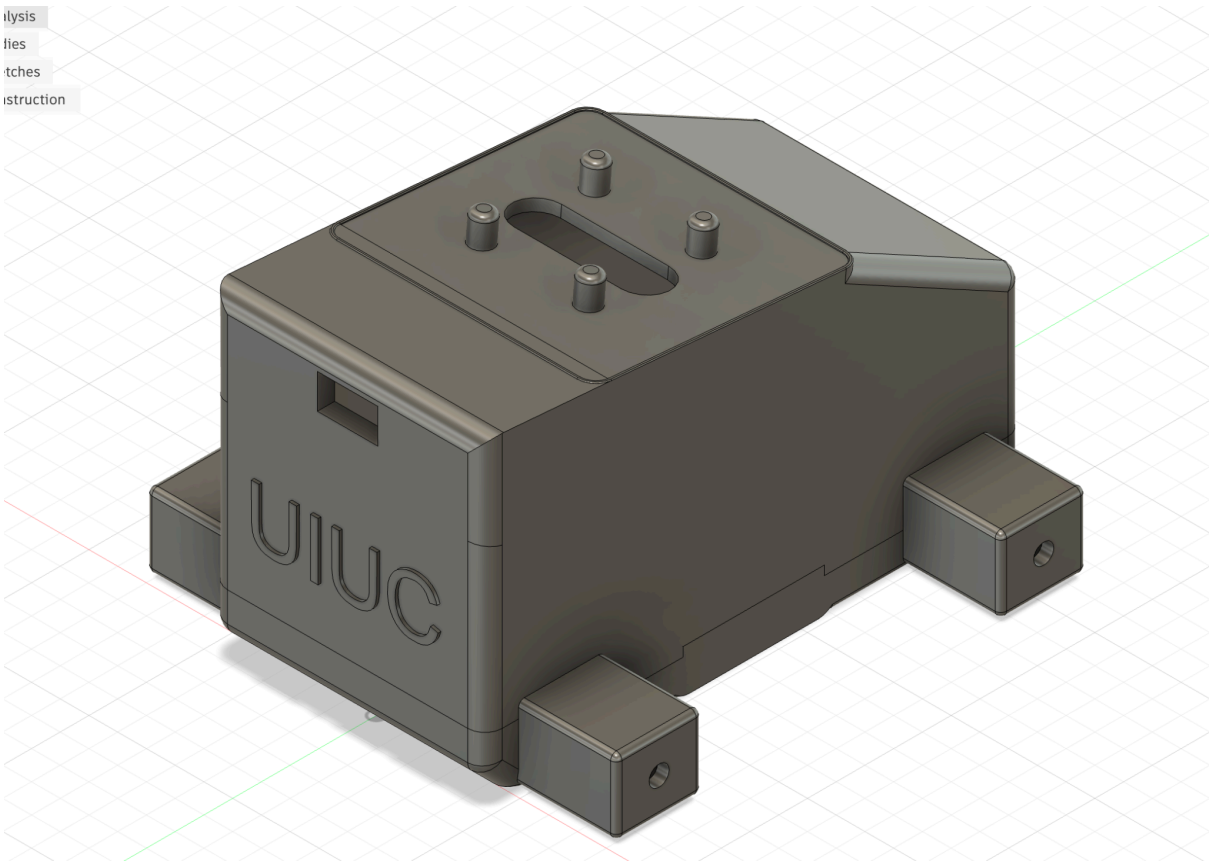
Challenges

- All modules needed to fit into a compact enclosure with very limited internal space
- The wiring and soldering had to remain stable after placing the components inside the robot
- The head and body both required careful arrangement to avoid interference between components
- Electrical connections needed to be checked carefully before moving into full functional testing

Next Steps

- Continue testing each module after full physical integration
- Verify the communication between the head PCB, body PCB, sensors, and interaction modules
- Test the robot's motion, sensing, display, microphone, and speaker functions as a complete system
- Improve wire management inside the enclosure if necessary
- Prepare for final system-level debugging and demonstration





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week 4.20 JIajun Gao
Weekly Progress Log – Week 4/20

Project: Interactive Desktop Companion Robot
Team: Jiajun Gao, Yuchen Shih, Zichao Wang

This Week's Progress

1. Web Controller for the Robot (Human-Control Mode)

We built and integrated a browser-based web controller for driving the robot over Wi-Fi, complementing the existing voice-driven `ai` mode. The implementation exercises the end-to-end path from a phone or laptop browser through the ESP32 WebSocket/link layer to motor commands, using the firmware's `human_control` mode (`robot.set_mode`) so manual driving does not conflict with wake-word and MCP-driven motion when modes are switched intentionally.

Work this week included wiring the controller UI to the same motion and safety stack used in normal operation (e.g., cliff and front IR behavior), and verifying that switching between voice control and web control is predictable for demo and lab use.

2. Voice Interaction System: Testing and Debug

We continued end-to-end testing of the voice pipeline: wake word, streaming audio to the local Mac backend, ASR → LLM → TTS, and state-driven behavior on the device. Sessions focused on reproducing intermittent issues (e.g., recognition under different mic conditions, timing between turns, and audio path stability) and tightening configuration so the stack is reliable for extended runs.

3. MCP Tools : Testing and Debug

We continued integration testing of Model Context Protocol tool calls from the LLM to the device (e.g., `robot.move`, `robot.get_status`, `robot.set_mode`, and related `self.*` tools). Debug work included confirming tool names/arguments match what the model emits, tracing failures in the dispatch path, and validating that motion and UI side effects match the user's intent after tool execution.

Key Achievements

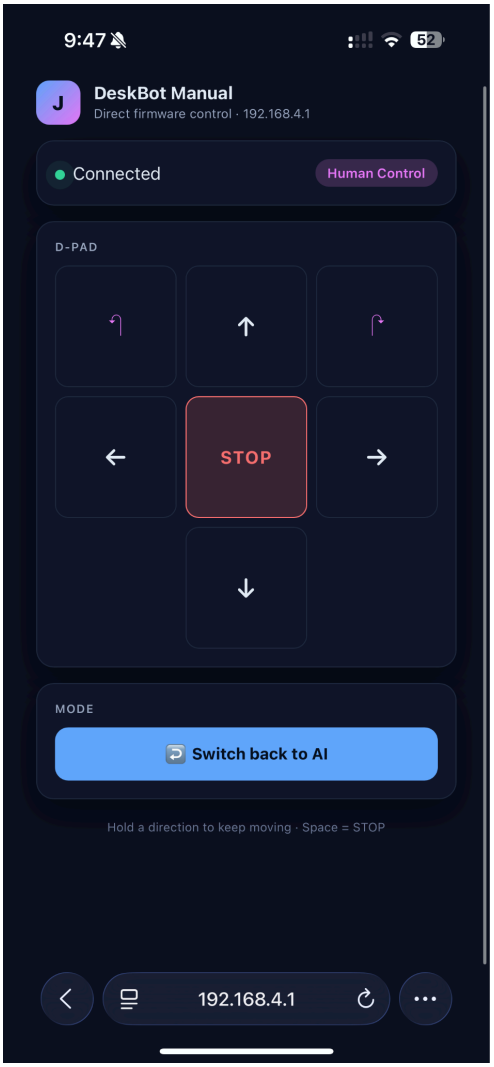
- Functional web-based controller for manual driving over Wi-Fi, aligned with `human_control` and existing safety logic
- Sustained testing of the full voice stack and MCP tool path on the integrated hardware + Mac server setup
- Clearer picture of failure modes and configuration limits for voice and tool use in real sessions

Challenges

- Balancing web UI latency and responsiveness with safe motor behavior at the edge of the desk
- Voice pipeline sensitivity to environment, power/mic quality, and network conditions
- MCP: occasional mismatches between natural-language requests and the exact tool schema the model must produce

Next Steps

- Harden the web controller UX (connection status, error feedback) and document a short operator checklist for mode switching
- Continue voice robustness work (e.g., ASR under battery vs. USB power, as noted in project status)
- Expand regression-style test passes for high-value MCP commands before milestone demos



week 4.27 Jiajun Gao
Weekly Progress Log – Week 4/27

Project: Interactive Desktop Companion Robot
Team: Jiajun Gao, Yuchen Shih, Zichao Wang

This Week's Progress

1. Web Panel

We extended the browser-side interface into a fuller web panel for operating the robot over Wi-Fi. Compared with the earlier minimal controller, the panel consolidates driving controls, connection/mode context, and operational feedback so we can run the robot from a laptop or phone during bench tests and demos without relying on the voice path.

The panel remains tied to the firmware's `human_control` path and the same motor and safety stack (cliff and front IR), so manual commands remain consistent with MCP-driven motion when switching modes.

2. Dance Function

We implemented and integrated a dedicated dance routine exposed to the stack as the `robot.dance` capability (voice/MCP and testing from the backend). The routine runs as a pre-programmed motion sequence on the order of tens of seconds, giving a concrete “personality” demo beyond simple directional moves.

Integration work included ensuring the dance does not fight safety locks (e.g., cliff latch), and that the face/UI state remains coherent while the sequence runs.

3. Latency Optimization

We measured and reduced perceived delay in the interactive loop—wake → streaming capture → server-side VAD/ASR → LLM → TTS → playback on device—and tightened whichever stages dominated in our setup (e.g., model/API turnaround, audio buffering, network/WebSocket behavior).

The goal was to improve responsiveness for short commands and MCP tool turns without sacrificing stability, so end-to-end times align better with our documented targets for direct commands and conversational turns.

Key Achievements

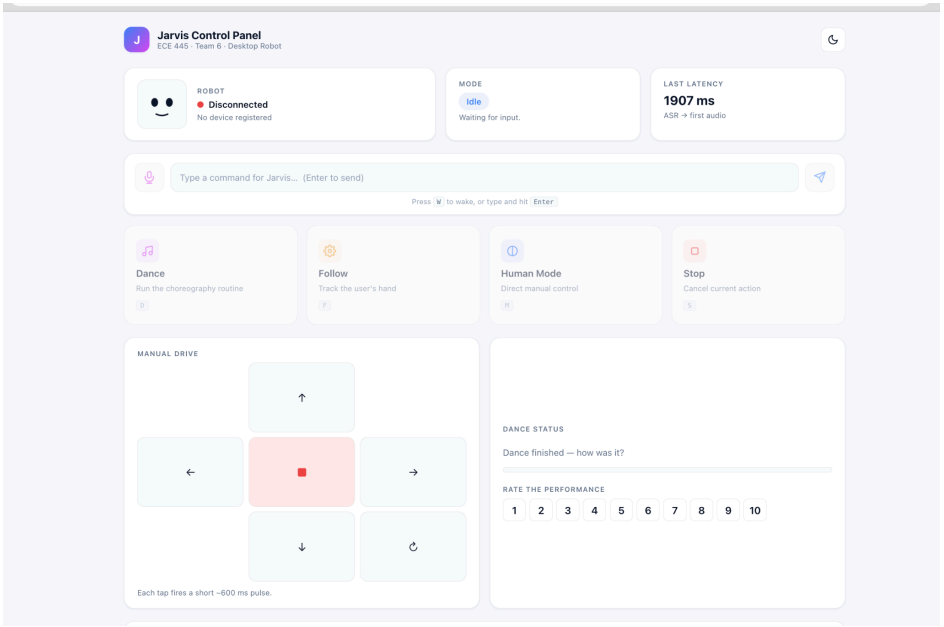
- Web panel suitable for repeatable manual control and demo operation alongside voice
- Working `robot.dance` behavior for showcase and regression testing
- Measurable latency improvements on the voice/MCP path for typical interactions

Challenges

- Web UI: clear connection and mode state so operators do not issue conflicting manual commands during voice or MCP activity
- Dance: sequencing duration vs. desk footprint and safety interrupts
- Latency: gains often trade off against robustness (buffer sizes, API choices); tuning required A/B style comparison on real hardware

Project Status

This week we completed the senior design project. The integrated voice stack, MCP-driven behavior (including dance), web panel, and latency tuning are finalized for demo and documentation. There is no further project plan beyond this deliverable.


Jarvis Control Panel
ECE 445 · Team 6 · Desktop Robot
ROBOT
Disconnected
No device registered
MODE
Idle
Waiting for input.
LAST LATENCY
1907 ms
ASR → first audio
Type a command for Jarvis... (Enter to send)
Press W to wake, or type and hit Enter
Dance
Run the choreography routine
D
Follow
Track the user's hand
F
Human Mode
Direct manual control
H
Stop
Cancel current action
S
MANUAL DRIVE
Each tap fires a short ~600 ms pulse.
DANCE STATUS
Dance finished — how was it?
RATE THE PERFORMANCE
1
2
3
4
5
6
7
8
9
10